

Challenge 3

Creating In-Game UI

Exam Objectives Covered

- Identify the process required to display data in various UI elements

Challenge Overview: Binding In-Game Data to UI Components

Welcome to the final challenge of your SkiFree-inspired game development course! Now that you've set up your functional title screen UI, it's time to shift your focus back to the main game UI. In this session, you will bind in-game data to your UI components so that the player can better understand the current state of the game.

Step 1: Add a UI Timer

Create a UI element that displays the current race time during the race.

Instructions:

1. Create Timer UI Element:
 - Add a text element to your in-game UI canvas to display the timer.
2. Bind Timer to Race Events:
 - Modify your RaceTimer script to update the timer UI element with the current race time.
 - Ensure the timer starts displaying when the race begins and updates in real-time as the player progresses.

Step 2: Display Leaderboard Scores

Create a UI component to display the top 5 race times and highlight the player's latest score if it makes the top 5.

Instructions:

1. Create Leaderboard UI:
 - Add a UI panel to your in-game UI canvas for the leaderboard.
 - Create text elements within the panel to display the top 5 race times.
2. Bind Leaderboard Data:

- Store the top 5 race times in a suitable data structure (e.g., a list or array).
- Update the leaderboard UI elements with the stored times when the race is over.

3. Highlight Latest Score:

- Implement logic to check if the player's latest race time makes it into the top 5.
- If the latest score is in the top 5, visually highlight it within the leaderboard UI (e.g., change text color or background).

Note: This session uses many of the same concepts covered in the first two sessions of this course. The primary difference is that instead of using the UI to trigger events, the UI will respond to in-game events.

Once you've completed these steps, your game will provide real-time feedback to the player through the UI, enhancing the overall gameplay experience.

Good luck with binding in-game data to your UI components! This exercise will solidify your understanding of UI interactions and data binding in Unity, preparing you for more advanced game development tasks. Happy developing!